Entropy-Degree Rigidity for Closed Orientable Non-Collapsed $\mathsf{RCD}(K, N)$ and Alexandrov Spaces

A Machine-Checked Formalization Blueprint in Lean 4

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Chapter 1

Metric-Measure Volume Entropy, Bishop–Gromov Growth, and Scaling

Definition 1.1 (Logarithmic ball-volume rate, exponential growth, and entropy-volume product)**.** For a metric-measure space $(Z, \mathsf{d}, \mathfrak{m})$ and basepoint $x_0 \in Z$, the logarithmic ball-volume rate is $R^{-1} \log \mathfrak{m}(B(x_0, R))$, exponential volume growth $\mathfrak{m}(B(x_0, R)) \le Ce^{hR}$ is captured by `HasExponentialVolumeGrowth`, the normalized exponential weights $w_s(x, z)$ normalize $e^{-s\,\mathsf{d}(x,z)}$ over finite samples, and the scale-invariant entropy-volume product is $h(\overline{X})^N \mathcal{H}^N(X)$.

Theorem 1.2 (Basepoint independence and Bishop–Gromov hyperbolic growth bound)**.** *By the triangle inclusion $B(x_0, R) \subseteq B(x_1, R + \mathsf{d}(x_0, x_1))$, exponential volume growth with exponent h transfers across all basepoints $x_0, x_1 \in Z$ and bounds $R^{-1} \log \mathfrak{m}(B(x_0, R)) \le h + R^{-1} \log C$. Moreover, the hyperbolic model density satisfies $(\sinh(\kappa r)/\kappa)^{N-1} \le (2\kappa)^{-(N-1)} e^{(N-1)\kappa r}$, giving the Bishop–Gromov upper bound $h(\overline{X}) \le (N-1)\kappa$ under $\mathsf{RCD}(-(N-1)\kappa^2, N)$.*

Proof. Triangle inequality ball monotonicity and the elementary estimate $2\sinh(u) \le e^u$ raised to the $(N-1)$-st power. □

Theorem 1.3 (Isometric invariance and π_1-equivariance of normalized exponential weights)**.** *Every distance-preserving action γ preserves $e^{-s\,\mathsf{d}(x,z)}$ and hence yields strictly positive, probability-normalized, $\pi_1(X)$-equivariant weights $w_s(\gamma x, \gamma z_i) = w_s(x, z_i)$.*

Proof. Direct invariance of $\mathsf{d}(\gamma x, \gamma z) = \mathsf{d}(x, z)$ inside the numerator and denominator sum. □

Theorem 1.4 (Homothetic invariance, normalization, and finite orientation-cover reduction)**.** *Under homothetic metric rescaling $(\mathsf{d}, \mathcal{H}^N) \mapsto (\lambda\mathsf{d}, \lambda^N \mathcal{H}^N)$ $(\lambda > 0)$, volume entropy scales by λ^{-1} and Hausdorff volume by λ^N, leaving $h(\overline{X})^N \mathcal{H}^N(X)$ invariant. Passing to the k-sheeted intermediate cover $\widehat{Y} \to Y$ associated to $f_*\pi_1(X) \le \pi_1(Y)$ with $|\deg(f)| = k|\deg(\hat{f})|$ and normalizing by $c = h(\overline{X})/h(\widetilde{Y})$ yields $|\deg(\hat{f})| = 1$, $c^{-1}h(\overline{X}) = h(\widetilde{Y})$, and $c^N \mathcal{H}^N(X) = \mathcal{H}^N(\widehat{Y})$.*

Proof. Cancellation of $\lambda^{-N}\lambda^N = 1$ and positive factor $k > 0$ in the covering degree factorization. □

Chapter 2

Sturm's CAT(0) Barycenter and Equivariant Natural Maps

Definition 2.1 (Sturm's renormalized d^2-barycenter functional and geodesic homotopy)**.** For a probability weight w supported at points y_i in a metric space (Y, d_Y) and reference basepoint $o \in Y$, Sturm's renormalized d^2-barycenter functional is $\mathcal{S}_{w,y,o}(z) = \sum_i w_i(\mathsf{d}_Y(y_i, z)^2 - \mathsf{d}_Y(o, y_i)^2)$. Together with the CAT(0) semi-parallelogram midpoint predicate `IsCATZeroMidpoint`, this supports the lifted geodesic homotopy $\overline{H}_s(x, t)$ and its quotient descent $H_s : X \times [0, 1] \to Y$.

Lemma 2.2 (First-moment linear growth bound and basepoint independence)**.** *Sturm's summand satisfies* $|\mathsf{d}(y, z)^2 - \mathsf{d}(o, y)^2| \le \mathsf{d}(o, z)(\mathsf{d}(o, z) + 2\,\mathsf{d}(o, y))$*, requiring only finite first moments, and shifting* $o_1 \mapsto o_2$ *adds a constant independent of* z*. Consequently,* $\mathcal{S}_{w,y,o}$ *is continuous and shares exact global minimizers with* `metricEnergy`*.*

Proof. Factor $a^2 - b^2 = (a - b)(a + b)$ with triangle inequality bounds and note that $\mathcal{S}_{w,y,o}(z) = 2\,\mathrm{metricEnergy}(w, y, z) - \sum_i w_i\,\mathsf{d}(o, y_i)^2$. □

Theorem 2.3 (CAT(0) midpoint deficit, Sturm variance inequality, and uniqueness)**.** *Under* `IsCATZeroMidpoint`*,* $\mathcal{S}_{w,y,o}$ *satisfies the strong midpoint convexity deficit* $\mathcal{S}_{w,y,o}(m) \le \frac{1}{2}\mathcal{S}_{w,y,o}(a) + \frac{1}{2}\mathcal{S}_{w,y,o}(b) - \frac{1}{4}\mathsf{d}_Y(a, b)^2$*. Any minimizer* $q = \mathrm{bar}(\nu)$ *therefore satisfies Sturm's variance inequality* $\frac{1}{2}\mathsf{d}_Y(q, z)^2 \le \mathcal{S}_{w,y,o}(z) - \mathcal{S}_{w,y,o}(q)$ *and is the unique global minimizer.*

Proof. Weighted summation of the CAT(0) semi-parallelogram law and comparison at $m = \mathrm{mid}(q, z)$ against minimality at q. □

Theorem 2.4 (π_1-equivariance of Sturm natural maps and geodesic homotopy descent (Lemma 3.1))**.** *For any isometric* $\pi_1(X)$*-representation* $\rho : \Gamma \curvearrowright \widetilde{Y}$*, unique minimization forces* $\mathrm{bar}(\rho(\gamma)_*\nu) = \rho(\gamma)\,\mathrm{bar}(\nu)$*. Combining with* Γ*-equivariance of* $w_s(x, z)$ *yields* ρ*-equivariance of the lifted natural map* $\overline{F}_s$ *and of the geodesic homotopy* $\overline{H}_s$*, which therefore descends to a homotopy* $H_s : X \times [0, 1] \to Y$ *between* f *(*$t = 0$*) and* F_s *(*$t = 1$*).*

Proof. Isometry covariance of $\mathcal{S}_{w,y,o}$ transfers the minimizing property of q to $\rho(\gamma)q$, and uniqueness yields equality. □

Chapter 3

Non-Collapsed $\mathsf{RCD}(K, N)$ Structure and CMMP Orientation Currents

Definition 3.1 (Non-collapsed $\mathsf{RCD}(K, N)$ stratification and Alexandrov specialization)**.** A non-collapsed $\mathsf{RCD}(K, N)$ space $(X, \mathsf{d}, \mathcal{H}^N)$ carries the regular set $\mathcal{R}_N(X)$ ($\theta_N(x) = 1$), the open topological manifold locus $\mathcal{M}(X) \supseteq \mathcal{R}_N(X)$, the closed singular set $\mathcal{S}(X) = X \setminus \mathcal{M}(X)$, and singular strata $\mathcal{S}^k(X)$. Closed Alexandrov N-spaces of curvature $\geq \kappa$ without boundary embed into closed non-collapsed $\mathsf{RCD}((N-1)\kappa, N)$ spaces via `AlexandrovSpaceWithoutBoundary`.

Theorem 3.2 (Codimension-two singular stratification and Bishop–Gromov density gap (Lemma 2.1 & Remark 4.3))**.** *On any closed non-collapsed $\mathsf{RCD}(K, N)$ space (and in particular on any closed Alexandrov N-space without boundary), $\mathcal{R}_N(X)$ and $\mathcal{M}(X)$ have full $\mathcal{H}^N$-measure, $\mathcal{S}(X) \subseteq X \setminus \mathcal{R}_N(X) \subseteq \mathcal{S}^{N-2}(X)$ is closed with $\mathcal{H}^{N-1}(\mathcal{S}(X)) = \mathcal{H}^{N-1}(X \setminus \mathcal{R}_N(X)) = 0$, $\mathcal{H}^N(X) > 0$, and every point $x \notin \mathcal{R}_N(X)$ satisfies the strict Bishop–Gromov density gap $\theta_N(x) < 1$.*

Proof. Outer-measure monotonicity along $\mathcal{S}(X) \subseteq X \setminus \mathcal{R}_N(X) \subseteq \mathcal{S}^{N-2}(X)$ and De Philippis–Gigli characterization $\theta_N(x) = 1 \iff x \in \mathcal{R}_N(X)$ with $\theta_N(x) \leq 1$. □

Definition 3.3 (Metric currents, fundamental manifold class, and CMMP orientation current)**.** `MetricCurrentOps` packages Ambrosio–Kirchheim / Lang metric currents with boundary ∂, mass measure $\|T\|$, total mass $\mathbf{M}(T)$, and Lipschitz pushforward $f_\#$. `FundamentalManifoldCurrent` captures the top-dimensional fundamental class $[\![Y]\!] \in \mathbf{I}_N(Y)$ on a closed oriented Riemannian N-manifold, while `CMMPOrientationCurrent` packages the Cavallucci–Marti–Mondino–Perales orientation current $T \in \mathbf{I}_N(X)$ ($\partial T = 0$, $\|T\| = \mathcal{H}^N$).

Theorem 3.4 (Mass and singular-nullity properties of the CMMP orientation current (Theorem 2.2))**.** *For the fundamental current $[\![Y]\!]$, $\mathbf{M}(d[\![Y]\!]) = |d|\,\mathcal{H}^N(Y)$ and $\mathcal{H}^N(Y) > 0$, making $d \mapsto d[\![Y]\!]$ injective on $\mathbb{Z}$. On X, the CMMP current T satisfies $\mathbf{M}(T) = \mathcal{H}^N(X) > 0$ and $\|T\|(\mathcal{S}(X)) = \|T\|(X \setminus \mathcal{R}_N(X)) = 0$.*

Proof. Follows from $\|T\| = \mathcal{H}^N$ combined with Theorem 3.2 and $\mathcal{H}^N(Y) > 0$. □

Chapter 4

Regular Values, Lipschitz Degree, and Homotopy Invariance

Definition 4.1 (Singular value image, regular value locus, and Lipschitz pushforward degree)**.** For a map $f : X \to Y$, $\Sigma_f := f(\mathcal{S}(X))$ is the singular value image, $V_f := Y \setminus \Sigma_f$ is the regular value locus, and $U_f := f^{-1}(V_f)$ is the regular domain locus. `LipschitzDegreePushforward` encodes $f_{\#}T = \deg(f)[\![Y]\!]$ via Federer constancy on Y together with the local degree integral formula on U_f.

Lemma 4.2 (Openness and full measure of regular values $V_f = Y \setminus f(\mathcal{S}(X))$ (Lemma 2.3))**.** *For any continuous map $f : X \to Y$ (X compact, Y Hausdorff), $U_f \subseteq \mathcal{M}(X)$, Σ_f is compact and closed, and V_f, U_f are open. When f is Lipschitz, $\mathcal{H}^{N-1}(\Sigma_f) = 0$ and $\mathcal{H}^N(V_f) = \mathcal{H}^N(Y)$.*

Proof. Continuous image of the closed compact set $\mathcal{S}(X)$ and Lipschitz Hausdorff bound $\mathcal{H}^{N-1}(f(\mathcal{S}(X))) \leq \mathrm{Lip}(f)^{N-1}\mathcal{H}^{N-1}(\mathcal{S}(X)) = 0$. □

Theorem 4.3 (Current pushforward degree identity and Lipschitz volume bound (Proposition 2.5))**.** *A Lipschitz map $f : X \to Y$ satisfies $\mathbf{M}(f_{\#}T) = |\deg(f)|\,\mathcal{H}^N(Y)$, $\deg(f) = \mathcal{H}^N(Y)^{-1}\int_Y \deg(f; U_X, y)\, d\mathcal{H}^N(y)$, and $|\deg(f)|\,\mathcal{H}^N(Y) \leq \mathrm{Lip}(f)^N \mathcal{H}^N(X)$ (in particular $\mathcal{H}^N(Y) \leq \mathcal{H}^N(X)$ when f is 1-Lipschitz and $\deg(f) \neq 0$).*

Proof. Combines $f_{\#}T = \deg(f)[\![Y]\!]$ with mass homogeneity $\mathbf{M}(f_{\#}T) \leq \mathrm{Lip}(f)^N \mathbf{M}(T)$. □

Theorem 4.4 (Wenger homotopy invariance, surjectivity, and homological index (Lemma 2.6, Cor. 2.8, Rmk. 4.2))**.** *Since $\partial T = 0$ and $\mathcal{H}^{N+1}(Y) = 0$, Wenger's homotopy formula forces $(f_1)_{\#}T = (f_0)_{\#}T$ and $\deg(f_1) = \deg(f_0)$ across Lipschitz homotopies. Any map with $\deg(f) \neq 0$ is surjective, degree multiplies across covers ($\deg(p \circ \hat{f}) = \deg(p)\deg(\hat{f})$), and Mitsuishi's homological index satisfies $\mathrm{ind}_H(f) \mid |\deg(f)|$ and $\mathrm{ind}_H(f) \leq |\deg(f)|$.*

Proof. Injectivity of $d \mapsto d[\![Y]\!]$ equates degrees from $(f_1)_{\#}T = (f_0)_{\#}T$; if range$(f)$ missed an open ball $U \subseteq Y$, positive measure of U would contradict support nullity of $\deg(f)[\![Y]\!]$. □

Chapter 5

Rectifiable Jacobians and $\mathsf{RCD}(K,N)$ Entropy-Degree Rigidity

Definition 5.1 (Metric Jacobian on Euclidean approximate tangent spaces (Lemma 3.2))**.** At $\mathcal{H}^N$-almost every regular point $x \in \mathcal{R}_N(X)$ where $\mathrm{Tan}(X,x) \cong E$ ($\dim_{\mathbb{R}} E = N$), the approximate linear differential $d_xF : E \to_l [\mathbb{R}]E$ defines the non-negative metric Jacobian $\mathrm{Jac}_x F := |\det(d_xF)|$.

Lemma 5.2 (Pushforward mass-Jacobian inequality (Lemma 3.3))**.** *Because each regular-fiber orientation sign satisfies* $|\sum_{x\in F^{-1}(y)\cap\mathcal{R}_N(X)} \mathrm{sign}(\det d_xF)| \le \#F^{-1}(y)$*, the rectifiable area formula yields*

$$|\deg(f)|\,\mathcal{H}^N(Y) = \mathbf{M}(F_{\#}T) \le \int_Y \mathrm{mult}_F(y)\,d\mathcal{H}^N(y) = \int_X \mathrm{Jac}_x F\,d\mathcal{H}^N(x).$$

Proof. Triangle inequality over finite fibers integrated against $\mathcal{H}^N_Y$, combined with Proposition 2.5. □

Theorem 5.3 (Pointwise entropy-Jacobian bound on $\mathcal{R}_N(X)$ (Proposition 3.4))**.** *On the Euclidean approximate tangent space* $\mathrm{Tan}(X,x) \cong \mathbb{R}^N$ *at* $x \in \mathcal{R}_N(X)$*, applying the rank-one Busemann Hessian equation and determinant inequality from* `BCG.RankOne` *yields* $\mathrm{Jac}_x F_s = |\det(d_xF_s)| \le (s/h(\widetilde{Y}))^N$.

Proof. Direct instantiation of `BCG.RankOne.abs_det_le_div_entropy_pow_of_rankOne_det_ineq`. □

Theorem 5.4 (Theorem 1.1 Part 1: Non-collapsed $\mathsf{RCD}(K,N)$ and Alexandrov entropy-degree inequality)**.** *For any closed orientable non-collapsed* $\mathsf{RCD}(K,N)$ *space* $(X,\mathsf{d},\mathcal{H}^N)$ *(or closed orientable* N*-dimensional Alexandrov space without boundary, Remark 4.3) and continuous map* $f : X \to Y$ *to a closed oriented negatively curved locally symmetric* N*-manifold* Y*,*

$$\mathrm{ind}_H(f)\,h(\widetilde{Y})^N\,\mathcal{H}^N(Y) \le |\deg(f)|\,h(\widetilde{Y})^N\,\mathcal{H}^N(Y) \le h(\overline{X})^N\,\mathcal{H}^N(X).$$

Proof. Combines Lemma 5.2 with Theorem 5.3 and passes to the limit $s \searrow h(\overline{X})$ via `BCG.bcg_volume_entropy_i` □

Theorem 5.5 (Theorem 1.1 Part 2: Mass-Jacobian sandwich and equality-case rigidity)**.** *When* $\deg(f) \neq 0$, $|\deg(f)| = [\pi_1(Y) : f_*\pi_1(X)] = k \geq 1$, *and equality holds in Theorem 1.1, homothetic rescaling by* $c = h(\overline{X})/h(\widetilde{Y})$ *and lifting to the degree-*k *cover* $p : \widehat{Y} \to Y$ *give* $|\deg(\hat{f})| = 1$ *and* $\mathcal{H}^N_c(X) = \mathcal{H}^N(\widehat{Y})$. *The* 1*-Lipschitz limit map* $\hat{F}_\infty : X \to \widehat{Y}$ *then saturates the mass-Jacobian sandwich (*$\mathrm{Jac}_x \hat{F}_\infty = 1$ *a.e.), is surjective, promotes to a global metric isometry* $(X, c\,\mathtt{d}_X) \xrightarrow{\sim} \widehat{Y}$, *and forces* $\mathcal{S}(X) = X \setminus \mathcal{R}_N(X) = \emptyset$.

Proof. Combines `finite_cover_entropy_degree_reduction`, `homothetic_normalization`, the a.e. integral sandwich $\mathcal{H}^N(\widehat{Y}) \leq \int_X \mathrm{Jac}_x \hat{F}_\infty \leq \mathcal{H}^N_c(X) = \mathcal{H}^N(\widehat{Y})$, surjectivity from $\deg(\hat{F}_\infty) \neq 0$, and Lipschitz-volume rigidity. □